

Primary Clinical Summary

Provider Reference Document — Multi-System Complex Case

Sarah K. Sahl • DOB: May 18, 1984 • Prepared June 2026

Purpose of This Document

This document was generated for provider review to assist in the management of a complex, multi-systemic presentation. All findings are sourced directly from clinical records, laboratory results, imaging reports, and confirmed diagnoses. This is the primary summary layer; the Deep Reference Appendix contains full raw data.

1. Patient Demographic Snapshot

Patient Name	Sarah K. Sahl
Date of Birth	May 18, 1984 (Age 41)
Gender	Female
Address	2623 NW Penny Ln, Ankeny, IA 50023-8355
Primary Contact	515-231-3688 (Mobile/Home)
Email	misssahl@outlook.com
Language	English (Spoken and Written)
Primary Insurer	Iowa Total Care
Known Allergies	NKAA (No Known Active Allergies) as of 05/29/2026

2. Critical Clinical Alerts & Contraindications

PHARMACOLOGICAL CONTRAINDICATION — NSAIDs STRICTLY PROHIBITED

NSAIDs (including meloxicam, ibuprofen, naproxen, and all related agents) are contraindicated. Meloxicam has been discontinued as of June 2026.

- Cardiac Risk:** Confirmed nonischemic cardiomyopathy with declining EF (56% → ~50%). NT-proBNP 263 pg/mL (above ref). Diastolic HTN 98 mmHg persisting on maximum beta blockade. NSAIDs increase cardiac preload and afterload.
- Renal Risk:** Documented eGFR fluctuations (73 in 2025). NSAIDs cause renal vasoconstriction and may precipitate acute kidney injury. Patient is on furosemide and potassium chloride — nephrotoxic agents must be strictly avoided.
- Confirmed Episode:** June 2026 — daily OTC ibuprofen use (bridge after meloxicam discontinuation without a replacement plan) produced acute bilateral ankle edema and BP elevation. Ibuprofen has been discontinued.

IV Contrast Caution

IV contrast should be used with caution given documented eGFR variability and concurrent nephrotoxic risk. Pre-procedure renal function verification recommended.

3. Genetic Findings & Genomic Data

3.1 Confirmed Pathogenic Variant — TTN

TTN c.73318_73324delGAATGCT (p.E24440MfsX2, heterozygous) — Pathogenic. A 7 base-pair frameshift deletion producing a truncated, non-functional titin protein. Titin haploinsufficiency causes the dilated cardiomyopathy phenotype. High family penetrance — father died from this disease, brother is further advanced in progression. Genetic anticipation pattern suspected across generations.

“I have ordered genetic consultation as I am concerned that in addition to the cardiomyopathy, there are musculoskeletal issues related to this genetic pathogenic variant.”

— Dr. Maria Poirier, M.D., Mayo Clinic — documented in clinical note

3.2 COL3A1 rs1800255 — Raw Genomic Data (Clinical Significance Pending)

Important: Research-Grade Data

Heterozygous COL3A1 rs1800255 (Ala698Thr) was identified in 23andMe raw genotype data. This is a vEDS-associated variant. Clinical significance for this patient is **pending formal genetics consultation** and has not yet been clinically validated. AlphaGenome computational modeling predicts a top 0.05% functional effect on COL3A1 expression in cardiac tissue (quantile 0.9995) — the highest signal in the full variant batch. This is a research prediction, not a clinical diagnostic finding.

Clinical correlation: COL3A1 encodes Type III collagen, the primary structural protein of blood vessels and hollow organs. The variant is cross-referenced with the documented vascular anomaly pattern (aberrant subclavian artery, apical blebs, redundant interatrial septum, multivalvular regurgitation). Formal genetics evaluation is listed as Priority 1 in Section 10.

3.3 Genetic Carrier Status

- Documented on Mayo Active Problem List as “Genetic Carrier of Other Disease” (noted 04/01/2026)
- AlphaGenome computational analysis performed on TTN deletion and 9 additional clinically relevant variants — outputs available in supplementary genomics documentation

4. Connective Tissue & Vascular Fragility

The following structural findings form a pattern consistent with a heritable connective tissue disorder. Each item is individually documented in imaging reports or physical examination notes. Their co-occurrence in a patient under 50 is clinically significant.

4.1 Vascular & Structural Anomalies

Finding	Detail	Source
Aberrant right subclavian artery	Retroesophageal origin from thoracic aorta	CT Chest 04/15/2026
Right apical lung blebs	Few right apical blebs present. In patients <50, strongly associated with heritable CTDs (Marfan, vEDS, hEDS)	CT Chest 04/15/2026
Redundant interatrial septum	Classic finding of floppy tissue in hypermobility disorders	Echocardiogram 07/2024
Multivalvular regurgitation	Mild mitral, tricuspid, and aortic regurgitation. Multi-valve involvement without rheumatic history is a CTD marker	Echocardiogram
Right paratracheal stripe prominence	CT recommended and performed for evaluation	CXR 04/01/2026

4.2 Hypermobility — Formally Documented

- Diagnosis: Hypermobility Joint — Active Problem, Mayo Clinic (noted 04/01/2026)
- Beighton markers confirmed by Dr. Maria Poirier on physical exam: knee hyperextension, ankle hyperextension, thumb-to-wrist apposition
- Chronic myalgia dating to high school — documented by Dr. Fernandes, UNMC Neurology (2020)

4.3 Tendon & Bone Pathology

Location	Finding	Source
Left shoulder	Partial bursal surface tearing of the supraspinatus tendon	MRI
Right hip	Trace edema/inflammation within greater trochanteric bursae. Gluteal tendons intact on Mayo re-read of outside MRI (10/20/2023)	MRI re-read 04/09/2026
Left iliac (pelvis)	7 mm hyperintense lesion	MRI
T3 vertebral body	9 mm sclerotic focus (nonspecific) + endplate degenerative changes in cervical spine	CT Chest 04/15/2026
Distal left femur	1.9 cm circumscribed sclerosis (nonossifying fibroma pattern)	MRI

Multi-site bone lesions: In the setting of chronic eosinophilia, these findings warrant consideration of eosinophilic bone marrow involvement, a recognized manifestation of hypereosinophilic syndrome. Orthopedic/radiology review to confirm T3 stability is listed as Priority 4 in Section 10.

5. Neurodivergent-Autonomic-Immune Overlap (AuDHD Triad)

This patient demonstrates a documented co-occurrence pattern involving neurodivergence, autonomic dysfunction, and immune/respiratory dysregulation — a triad recognized in the literature in the context of mast cell activation, autonomic dysfunction, and connective tissue disorders.

5.1 Neuropsychiatric

- ADHD — confirmed; managed with Atomoxetine (Strattera)
- OCD — confirmed diagnosis
- Prolonged PTSD — confirmed; FKBP5 stress-response variants present on genomic analysis
- Anxiety — managed with buspirone and alprazolam XR

5.2 Autonomic / Cardiac

- Inappropriate Sinus Tachycardia (IST) — confirmed diagnosis; pre-treatment resting HR 90s overnight, 120-130 bpm daytime (documented UNMC Cardiology)
- Tachycardia — active diagnosis
- Symptoms: dizziness, lightheadedness, profuse sweating consistent with dysautonomia
- 2018 catecholamine testing: markedly elevated isolated dopamine (autonomic dysregulation pattern) — not yet repeated for comparison

5.3 Immune / Mast Cell / Allergic

- Idiopathic urticaria (chronic hives) — documented from at least 2011; hematology notes explicitly raised mast cell involvement in context of eosinophilia
- Tryptase 2.3 ng/mL (normal, <11.5) — systemic mastocytosis excluded April 2026
- Severe environmental allergies: pollen, ragweed, dust mites
- Severe Persistent Asthma — eosinophilic phenotype; FeNO 82 ppb (2× ULN)

6. Pulmonary & Hematological Profile

6.1 Respiratory Diagnostics — 04/15/2026

Test	Pre-Bronchodilator	Post-Bronchodilator	Flag
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FeNO (Exhaled Nitric Oxide)	82 ppb	N/A	HIGH — ULN 39 ppb
FEV1/FVC	57.39%	73.70%	Obstructive; sig BD response (+16 pts)
FVC (L)	3.65	4.06	
FEV1 (L)	2.10	2.99	
FEF 25-75% (L/s)	1.00	2.21	Significant small airways improvement
DLCO (ml/min/mmHg)	17.68	N/A	
TLC (L)	5.83	N/A	Air trapping context
RV (L)	2.13	N/A	

FeNO 82 ppb: Markedly elevated — more than double ULN. The FeNO/eosinophilia pairing strongly confirms a Type-2 (Th2-high) inflammatory phenotype. This provided the clinical rationale for mepolizumab initiation. Concurrent medications at time of testing: Albuterol, Advair, Singulair.

6.2 Hematological Findings — 04/09/2026

Component	Value	Reference Range	Status
Hemoglobin	12.3 g/dL	11.6–15.0	Normal
Leukocytes (WBC)	4.4 x10(9)/L	3.4–9.6	Normal
Neutrophils (auto)	1.51 x10(9)/L	1.56–6.45	Low (L)
Neutrophils (manual ANC)	1.85 x10(9)/L	1.56–6.45	Normal — see note
Lymphocytes	1.90 x10(9)/L	0.95–3.07	Normal
Eosinophils	0.58 x10(9)/L	0.03–0.48	HIGH (H)
Eosinophils % (manual smear)	12%	1–3%	HIGH (H)
Basophils	0.06 x10(9)/L	0.01–0.08	Normal
Basophils % (manual smear)	3%	0–2%	HIGH (H) on %
Tryptase	2.3 ng/mL	<11.5	Normal

Neutrophil discrepancy note: Automated count 1.51 (Low) vs manual ANC 1.85 (Normal). Discrepancy between automated and manual counts — manual differential is more reliable for accurate differential classification in the setting of active eosinophilia.

6.3 Exclusion Workup — All Negative (Mayo Clinic April 2026)

All of the following conditions have been formally excluded:

- **ANCA-positive vasculitis:** MPO <0.2, PR3 <0.2 — Negative
- **Systemic mastocytosis:** Tryptase 2.3 ng/mL (ref <11.5) — Normal
- **B-cell lymphoproliferative disorder:** Flow cytometry: no monocytic B-cell population
- **T-cell lymphoproliferative disorder:** TCR PCR: no clonal T-cell rearrangement
- **Autoimmune CTD:** ANA <1:80 — Negative
- **FIP1L1-PDGFRα positive CEL:** FISH panel: all loci normal (4q12, 5q32, 8p11.2, 9p24.1, 9q34, 13q12.2)
- **Parasitic infection:** GI pathogen panel (PCR) and O&P microscopy: all negative

7. Cardiovascular Status

7.1 Active Diagnoses

- Nonischemic Cardiomyopathy (TTN pathogenic variant) — EF trend 56% → ~50%
- Cardiomyopathy Family History — father deceased from disease; brother advanced progression
- Inappropriate Sinus Tachycardia — managed with metoprolol succinate 100mg BID
- Tachycardia — active diagnosis

7.2 Recent Cardiac Data

ECG — 04/01/2026	HR 59 bpm, PR 148 ms, QTc 411 ms, QRSD 78 ms — Normal intervals
Vitals — 04/16/2026	Pulse 74 bpm, SpO2 98%
BP — 04/08/2026	109/75 mmHg (note: diastolic HTN 98 documented on max beta blockade in separate encounter)
NT-proBNP — 04/01/2026	263 pg/mL — FLAGGED ABOVE REF (≤162). Mayo noted below acute HF triage threshold (300) but did not pursue. No prior baseline in record for trending.
NT-proBNP — 02/08/2026	4 pg/mL — Normal (Broadlawns ED)

NT-proBNP framing: 263 pg/mL is above the chronic cardiomyopathy monitoring reference (≤162), obtained before a subsequent acute fluid retention episode, with no prior baseline for trend comparison. The relevant clinical question is not “is this acute CHF” but “what is this patient’s NT-proBNP trajectory.” Repeat draw recommended.

8. Urological & Radiological Findings

8.1 Urological Profile

- CT Urogram (04/01/2026): 3 mm filling defect with probable enhancement in posterior base of bladder. Stable and unchanged vs 10/13/2016 and 12/23/2025.
- Cystoscopy (04/03/2026): Finalized. Bladder filling defect requires ongoing correlation.
- Active Problem List: Trigonitis Bladder (noted 05/01/2026) and Mass Bladder (noted 04/01/2026)
- Urinalysis (04/01/2026): WBC 1-3/hpf — automated flag “Abnormal” but within female-specific reference range (<11/hpf); clinically normal finding
- Bacterial culture: Urogenital microbiota only; susceptibilities not performed

8.2 Ancillary CT Urogram Findings

- Tiny hepatic cysts/hemangiomas — incidental finding
- Splenule — incidental finding
- No urachal remnant or cyst identified

8.3 Skeletal Imaging

- T3 vertebral body: 9 mm sclerotic focus (nonspecific) — orthopedic/radiology review for stability listed as Priority 4
- Cervical spine: endplate degenerative changes
- Left iliac: 7 mm hyperintense lesion (see Section 4.3)
- Distal left femur: 1.9 cm circumscribed sclerosis (see Section 4.3)

9. Current Pharmacotherapy & Interventions

Meloxicam is DISCONTINUED as of June 2026. It appears on some older records as active — this is a records lag. NSAIDs are strictly contraindicated (see Section 2). Current bridge gap: pain management appointment June 18, 2026.

Medication	Dose	Started	Notes
Mepolizumab (Nucala)	100mg SQ monthly	04/16/2026	Anti-IL-5 biologic. Initiated Mayo Clinic. First-line for eosinophilic asthma. NEW.
Metoprolol succinate	100mg BID	01/03/2026	Beta-1 blocker for IST + cardiomyopathy. ADRB1 rs1801253 variant — receptor may respond atypically.
Fluticasone-salmeterol (Advair)	250-50 mcg 1 puff BID	02/08/2026	ICS+LABA. ADRB2 variant affects salmeterol target receptor. GLCC11 variant predicts reduced steroid response.
Gabapentin (Neurontin)	400mg 4x daily	01/07/2020	1,600mg/day total. CURRENT — UNDER REVIEW at pain management 06/18/2026. Pregabalin being considered as replacement.

Baclofen (Lioresal)	10mg daily PRN	03/09/2020	Muscle relaxant for spasms. Also under review at pain management 06/18/2026.
Furosemide	20mg PRN	07/08/2024	Loop diuretic. Refill pending as of June 2026. Active edema episode.
Potassium Chloride	10 mEq with furosemide	07/08/2024	Electrolyte replacement with furosemide use.
Tramadol (Ultram)	50mg q6h PRN	02/19/2018	Analgesic. OPRM1 A118G homozygous AA — altered opioid receptor binding. CYP2D6 *6 allele affects metabolism.
Rosuvastatin (Crestor)	5mg daily	12/05/2025	Statin. SLCO1B1 *5 allele — increased myopathy risk. Monitor myalgia (confounded by baseline MSK pain).
Fluoxetine (Prozac)	10mg daily	02/26/2026	SSRI. CYP2C19 *2 poor metabolizer (homozygous AA) — drug accumulation risk.
Buspirone (BuSpar)	5mg BID	03/14/2026	Anxiolytic.
Alprazolam XR (Xanax XR)	0.5mg daily	—	Benzodiazepine anxiolytic.
Montelukast (Singulair)	10mg daily (bedtime)	09/10/2024	Leukotriene receptor antagonist for asthma.
Albuterol	90mcg/actuation 2 puffs PRN	—	Rescue bronchodilator. ADRB2 variant affects target receptor.
Estradiol (Climara)	0.1mg/24hr patch 2x weekly	—	HRT. Insurance note: Iowa Total Care will only fill 1-month supply. Must be written as 1-month Rx. Amazon Pharmacy preferred.
Cetirizine (Zyrtec)	10mg daily	—	Antihistamine for allergic symptoms/urticaria.
Atomoxetine (Strattera)	Active	—	Non-stimulant ADHD. CYP2D6 metabolized — *6 allele may affect levels.
Acetaminophen	1,000mg PRN	—	Currently primary analgesic (ibuprofen discontinued). Insufficient for pain control — bridge gap until 06/18/2026.

9.1 Analgesic Protocol Note

- Indicated: Acetaminophen 1,000mg PRN (currently primary analgesic — insufficient)
- Indicated PRN: Tramadol 50mg q6h PRN
- CONTRAINDICATED: All NSAIDs including meloxicam, ibuprofen, naproxen
- Under review at 06/18/2026 pain management: gabapentin continuation vs. pregabalin; baclofen vs. more targeted muscle relaxant for CTD-related spasm

10. Provider Review Priorities

#	Priority	Detail
1	Genetics Consultation	Clinical interpretation of COL3A1 rs1800255 (heterozygous) is required to determine potential vEDS or other vascular fragility associations. Cross-reference with vascular anomaly pattern (aberrant subclavian, apical blebs, multivalvular regurgitation).
2	Urological Follow-up	Cystoscopy finalized 04/03/2026. Ongoing evaluation of 3mm bladder filling defect in context of Trigonitis (05/01/2026) diagnosis. Stable since 2016.
3	Renal Protection	Continued eGFR monitoring. Strict avoidance of all nephrotoxic agents (NSAIDs, IV contrast caution). Furosemide + potassium chloride on board.
4	T3 Sclerotic Focus	9mm sclerotic focus in T3 vertebral body. Orthopedic/radiology review to confirm stability or need for bone density studies.
5	NT-proBNP Repeat	263 pg/mL (above ref ≤162) obtained 04/01/2026. No prior baseline. Repeat draw now recommended to establish trend in context of recent fluid retention episode.
6	Pain Management — 06/18/2026	Bridge gap from meloxicam discontinuation. Gabapentin under review (pregabalin preferred). Baclofen review. COMT Val158Met variant (intermediate pain sensitivity) relevant to dosing.
7	Multi-System Coordination	Cardiology (IST/Cardiomyopathy/TTN), Hematology/Immunology (EGPA vs HES differential), Pulmonology (mepolizumab titration), Genetics (CTD evaluation), Pain Management — all require coordinated care architecture.

All findings sourced from clinical records • Prepared June 2026 • Not a self-diagnosis